

UPC/EAN and Code 39 Barcode Generator

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Presented here is a program that generates both UPC/EAN/IAN and Code 39 barcodes using JavaScript.

Barcode is an optical machine-readable representation of data relating to the object to which it is attached. Barcodes are of various types and are used in various places. Earlier, barcode was a one-dimensional (1D) code of variable sizes, used to encode information that can be directly fed to a computer terminal with the help of barcode readers, without any physical contact with the item. Now you can

also find two-dimensional (2D) codes used in many places.

Matrix code, also termed as 2D barcode or simply 2D code, is a two-dimensional way to represent information. These codes have information encoded in both horizontal and vertical directions. Quick response (QR) code is one of the examples of 2D codes. (For more details about QR code, please refer to a software article published in EFY Sep. 2012 issue.)

2D barcode is similar to linear or 1D barcode but can represent more data per unit area.

UPC/EAN/IAN barcode

Universal Product Code (UPC), European Article Number (EAN) and International Article Number (IAN) are 1D barcodes. These codes have information encoded in only one horizontal direction; there is no information in the vertical direction.

UPC barcode. There are different types of UPC codes including UPC-A and UPC-E codes. Each UPC-A barcode consists of a scannable strip of black bars and white spaces, above a sequence of twelve numerical digits. A typical UPC barcode for '12345678999' is shown in Fig. 1.



Fig. 1: Typical UPC barcode

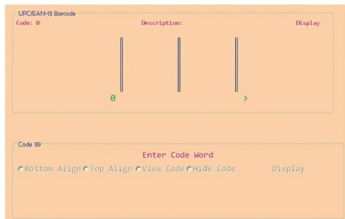


Fig. 2: Program output



of the original 12-digit UPC system developed in the United States. EAN-13 barcode is defined by the standards organisation GS1.

EAN-13 barcodes are used worldwide for marking products often sold at retail point-of-sale. The numbers encoded in EAN-13 barcodes are product identification numbers. All the numbers encoded in UPC and EAN barcodes are known as Global Trade Item Numbers (GTINs), and they can be encoded in other GS1 barcodes.

Structure of EAN-13. EAN-13 barcode is a numeric-only code, divided into three parts: first six digits come under one slab, next six digits come under another slab and the last digit is the parity digit. The first digit is placed before the barcode and in the last '>' symbol is used to notify the scanner that it is the last digit, so that the scanner can end the scanning process and begin processing the data bars.

Code 39 barcode

Code 39 is a general-purpose code widely used worldwide. It is also one of the examples of 1D barcode. Code 39 is widely used in many industries and is the standard for many government barcode specifications, including the US Department of Defense. It is defined in American National Standards Institute (ANSI) standard MH10.8M-1983.

Structure. Code 39 is an alphanumeric bar code that starts and ends with an asterisk (*). This barcode was used to encode data of various articles and packages till recently.

More information on these barcodes can be found on http://en.wikipedia.org/wiki/Code_39 and http://en.wikipedia.org/wiki/Universal_Product_Code_websites.